MINISTRY OF NEW AND RENEWABLE ENERGY
GOVERNMENT OF INDIA

Bharat RE Summit & Expo 2026
Invest · Innovate · Inspire

RENEWABLE ENERGY RESEARCH POSTER CHALLENGE 2026

Abstract Submission

ShowcasingResearchandInnovationforIndia'sCleanEnergy Future

Extraction and Utilisation of Lignin as Bio-Additive in a Compression Ignition Engine Fuelled with Diesel Fuel – An Approach to Improve Fuel Properties

Srinath Vatchavari Venkateshan, Aarya Brat Shankar and R.Bhuvaneshwaran

[1]Department OF Mechanical Engineering, KARE University,Krishnankoil, Tamil Nadu India;e-mail: 9926009061@klu.ac.in| [2]Department OF Bio Technology , KARE University,Krishnankoil,Tamil Nadu, India | [3]Department OF Electrical and Electronics Engineering (EEE) , KARE University, Krishnankoil, Tamil Nadu, India

Presenting Author: Srinath Vatchavari Venkateshan **Thematic Area:** Bioenergy & Waste-to-energy

Background / Research Problem : Diesel engines depend heavily on petroleum-based fuels, creating a need to explore alternative fuel additives. Lignin, a carbon-rich component of biomass, offers potential for conversion into liquid products through pyrolysis. However, the suitability of lignin-derived bio-additives for diesel blending requires evaluation of their fuel properties, combustion behaviour and engine emissions.

Objective / Research Question : To extract lignin from suitable biomass, produce bio-additives through controlled pyrolysis, and evaluate their physicochemical properties and suitability for blending with diesel at different dosage levels.

Key Findings / Results : The study will investigate lignin extraction from suitable biomass and the production of bio-additives through controlled pyrolysis. The resulting liquid products will be characterized for density, viscosity, calorific value, flash point and chemical composition. Diesel blends containing different bio-additive proportions, up to a maximum of 20%, will be evaluated for fuel properties, engine performance, combustion characteristics and emissions. Experimental comparisons with conventional diesel will help assess blend suitability and identify a promising dosage level.

Novelty and Potential Application : This study integrates lignin extraction from biomass, controlled pyrolysis, bio-additive characterization and diesel-blend evaluation in a compression-ignition engine. It explores the potential to valorize biomass resources and partially substitute petroleum-based diesel. Further optimization and pilot-scale assessment could help determine its technical feasibility, scalability and contribution to India's renewable energy development.

Keywords: Lignin, Bio-additives, Pyrolysis, Diesel Blending, Compression-Ignition Engine.

References: [1] H. Yang, R. Yan, H. Chen, D. H. Lee, and C. Zheng, "Characteristics of hemicellulose, cellulose and lignin pyrolysis," Fuel, vol. 86, no. 12–13, pp. 1781–1788, 2007. https://doi.org/10.1016/j.fuel.2006.12.013

Acknowledgements : The authors acknowledge Kalasalingam Academy of Research and Education (KARE) for its institutional support and express their sincere gratitude to their mentor, Dr. S. K. Gowthaman, for his guidance and encouragement throughout the research project.

Graphical Abstract :

